

Revised Generation Interconnection System Impact Study Report

For Queue Project Q19-1237

Dg Amp Solar Coldwater 138 kV
September 20, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1237. The Interconnection Customer proposes a 237.7 MW battery energy storage facility interconnecting at the Dg Amp Solar Coldwater 138 kV substation in the Virginia area. The request was received on July 07, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 436 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$51,534,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Kings River – S Meadow 2 138kV
Contingency: Centralia – Highland City 138kV
Loading: 175.9% (Rating: 239 MVA)
DFAX: 100.0%
Pre-project loading: 76.5%
2. Facility: Kings River – Dg Amp Solar Coldwater 138kV line
Contingency: Highland City – Centralia 138kV
Loading: 174.3% (Rating: 217 MVA)
DFAX: 100.0%
Pre-project loading: 64.7%
3. Facility: Westrock Southeast Llc – Pacifica 138kV
Contingency: Kings River – South Meadow 2 138kV L
Loading: 169.2% (Rating: 195 MVA)
DFAX: 100.0%
Pre-project loading: 47.0%
4. Facility: Pacifica – Mukonago 138kV
Contingency: Kings Riv – S Meadow 2 138kV
Loading: 169.2% (Rating: 222 MVA)
DFAX: 100.0%
Pre-project loading: 62.3%
5. Facility: Highland City – Mukonago 138kV
Contingency: South Meadow 2 Substation – Kings Riv 138kV
Loading: 168.0% (Rating: 235 MVA)
DFAX: 100.0%
Pre-project loading: 66.8%

6. Facility: Westrock Southeast Llc – Dg Amp Solar Coldwater 138kV
Contingency: Kings Riv – South Meadow 2 138kV
Loading: 158.9% (Rating: 193 MVA)
DFAX: 100.0%
Pre-project loading: 35.7%
7. Facility: Centrlia – Highland City Substation 138kV
Contingency: Kings River – S Meadow 2 138kV Line
Loading: 140.6% (Rating: 299 MVA)
DFAX: 100.0%
Pre-project loading: 61.2%
8. Facility: Jim Hill – Cedar Bayou Plant 138kV
Contingency: South Meadow 2 – Kings River 138kV
Loading: 129.5% (Rating: 236 MVA)
DFAX: 59.1%
Pre-project loading: 70.0%
9. Facility: Cedar Bayou Plant – Centralia 138kV line
Contingency: Kings Riv – S Meadow 2 138kV L
Loading: 126.7% (Rating: 208 MVA)
DFAX: 59.1%
Pre-project loading: 59.2%
10. Facility: Dps Ocean Spray – Four Rivers 138kV
Contingency: Birchwood – Leavitt 345kV
Loading: 118.4% (Rating: 242 MVA)
DFAX: 8.3%
Pre-project loading: 110.3%
11. Facility: Yellow reek – Walthall 138kV
Contingency: N Tower Tap – Unknown 138kV
Loading: 112.8% (Rating: 247 MVA)
DFAX: 7.9%
Pre-project loading: 105.1%
12. Facility: Leavitt – S Meadow 2 138kV Ckt 1
Contingency: S Meadow 2 – Leavitt 138kV Ckt 2
Loading: 111.8% (Rating: 464 MVA)
DFAX: 50.2%
Pre-project loading: 86.0%
13. Facility: Leavitt – S Meadow 2 138kV Cir. 2
Contingency: S Meadow 2 – Leavitt 138kV Circuit 1
Loading: 111.8% (Rating: 464 MVA)
DFAX: 50.2%
Pre-project loading: 86.0%
14. Facility: Haggin Substation – Jim Hill 138kV line
Contingency: Kings Riv – S Meadow 2 138kV
Loading: 110.6% (Rating: 324 MVA)
DFAX: 59.1%
Pre-project loading: 67.4%
15. Facility: Mitchell Dam – Skaggs Island Junction 2 345kV
Contingency: Leavtt – Chancellor Row 345kV
Loading: 109.6% (Rating: 1,191 MVA)
DFAX: 11.7%
Pre-project loading: 107.2%
16. Facility: Shack Mills Sta – Chancellor Row 138kV
Contingency: Chancellor Row – Bear Island 345kV

Loading: 107.5% (Rating: 229 MVA)

DFAX: 5.6%

Pre-project loading: 101.7%

17. Facility: Haggin – Walthall 138kV

Contingency: Haggin – North Tower Tap Substation 345kV L

Loading: 102.8% (Rating: 563 MVA)

DFAX: 11.0%

Pre-project loading: 98.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kings Riv – S Meadow 2 138kV	Rebuild line with higher-capacity conductor	\$7,585,000	100% (\$7,585,000)
Kings River – Dg Amp Solar Coldwater 138kV line	Rebuild line with higher-capacity conductor	\$5,244,000	100% (\$5,244,000)
Westrock Southeast Llc – Pacifica 138kV	Rebuild line with higher-capacity conductor	\$5,325,000	100% (\$5,325,000)
Pacifica Substation – Mukonago 138kV line	Rebuild line with higher-capacity conductor	\$4,088,000	100% (\$4,088,000)
Mukonago – Highland City 138kV L	Rebuild line with higher-capacity conductor	\$13,719,000	100% (\$13,719,000)
Westrock Southeast Llc – Dg Amp Solar Coldwater 138kV	Rebuild line with higher-capacity conductor	\$2,038,000	100% (\$2,038,000)
Centrlia – Highland City Substation 138kV	Rebuild line with higher-capacity conductor	\$5,392,000	100% (\$5,392,000)
Jim Hill – Cedar Bayou Plant 138kV	Rebuild line with higher-capacity conductor	\$3,667,000	56% (\$2,056,000)
Centralia – Cedar Bayou Plant 138kV	Rebuild line with higher-capacity conductor	\$1,496,000	56% (\$839,000)
Dps Ocean Spray – Four Rivers 138kV	Rebuild line with higher-capacity conductor	\$7,468,000	26% (\$1,918,000)
Yellow reek – Walthall 138kV	Rebuild line with higher-capacity conductor	\$3,060,000	15% (\$466,000)
Leavitt – S Meadow 2 138kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,326,000	67% (\$884,000)
Leavitt – S Meadow 2 138kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,218,000	67% (\$812,000)
Haggin Substation – Jim Hill 138kV line	Rebuild line with higher-capacity conductor	\$1,101,000	56% (\$618,000)
Mitchell Dam – Skaggs Island Jct 2 345kV	Reconductor line	\$19,021,000	2% (\$352,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shack Mills Sta – Chancellor Row 138kV	Reconductor line	\$829,000	11% (\$89,000)
Haggin – Walthall 138kV	Reconductor line	\$1,726,000	6% (\$109,000)
	Total allocated to project		\$51,534,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.